Dynamic Web Extended- DWE User Guide

This is a guide to report creators on how to use the *Dynamic Web Extended* addon to generate website pages from a Gramps family tree. It is also a guide for report visitors that describes what they will see and how to use the website.

For Gramps 5.2 and 6.0

Revised October 2026, John Vitlin

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1 Introduction

Dynamic Web Extended (called **DWE** for short) is a Gramps report addon for creating interactive genealogy websites. It is derived from the Gramps **Dynamic Web Report** (**DWR** for short), originally developed by Pierre Bélissent and others, and retains the main features and layout of that report while adding several extensions.

DWE is intended for family-history websites and is useful to explore not only the details recorded for an individual, but also that person's place within their wider family.

The principal additions include:

- **Events**, which has been extended and combines a person's own events with relevant events from parents, siblings, spouses and children to provide a chronological view of the family around them.
- **Lifespans**, a graphical display showing how the lives of a person and their immediate family overlap.
- **Tree Viewer**, which provides an interactive graphical view of the family tree using a bundled version of the Topola Viewer.
- Additional ways of explaining a person's place in the family tree, including the **Tracking**, **Why Me**, and **Story** features, and clearer handling of people belonging to multiple families.
- **Markdown notes**, allowing selected Gramps notes written in Markdown to be rendered as formatted content.
- **Local Labels**, allowing selected labels used in the generated website to be customised.
- Extra **Website theme**s to change the look of the report. These can be selected from within the website.
- Improvements to displaying relationships, place names and other information.

DWE does not replace the standard *Dynamic Web Report*. It is a separate addon, so both reports can be installed and used in Gramps.

This guide concentrates on the features added or changed by DWE. For the underlying features inherited unchanged from *Dynamic Web Report*, the standard Gramps *Dynamic Web Report* documentation remains applicable. That documentation is online on the Gramps website.

1.1 About this guide

Features inherited unchanged from the standard Gramps *Dynamic Web Report* are not described in detail.

DWE has been tested with **Gramps 5.2** and **Gramps 6.0**. The test operating system was Windows 10. The screenshots in this guide may differ slightly between Gramps versions, but the operation of DWE is the same. Most of the screenshots use the Mainz website template.

The web pages of the report can be displayed in different formats at any time, determined by a visitor via a **Configuration** option. The configuration button (shown as a gear icon) is at the top right of the report's title bar. The **Use tabbed panels instead of sections** option determines whether the pages are displayed in tabbed panels or as linear sections. The screenshots in the guide assume the website user has selected the tabbed panels view, but the information in this guide applies to all views.

1.2 Report and website terminology

Gramps uses the term **report** for the *Dynamic Web Report*. *Dynamic Web Extended* is similar and is selected from the Gramps **Reports** menu. Running the report generates a set of web pages that are viewed in a web browser.

In this guide, **DWE report** generally refers to the report being created in Gramps, while **DWE website** refers to the generated result viewed in a browser. They are two ways of referring to the same generated family-history material.

2 History

Dynamic Web Extended began as a personal modification of the Gramps *Dynamic Web Report* (DWR), originally developed by Pierre Bélissent and others. The word 'Dynamic' can indicate that the website is configurable - site visitors can change the look by using Configuration options. 'Dynamic' also indicates that each page is prepared dynamically in the browser from data files as required; the pages are not individual static pages.

The first version was called **Dynamic Web Report B (DWR B)**. It began with changes to the Events display and gradually acquired additional features intended to make it easier to understand people in the context of their families.

Development continued from time to time. The family timeline developed into the current **Lifespans** visualisation, Markdown note support was added, and the Topola Viewer was integrated and bundled with the addon to provide the **Tree Viewer**. Other changes improved events, relationships, historical place names, translations and local viewing.

Because the addon became substantially more than a modified copy of *Dynamic Web Report*, it was reorganised and renamed **Dynamic Web Extended (DWE)**. The new name reflects its purpose: retaining the foundation and much of the functionality of *Dynamic Web Report* while extending it with additional ways of presenting and exploring family-history information.

3 Installing Dynamic Web Extended

Dynamic Web Extended is distributed as a Gramps addon. It uses the standard Gramps addon distribution and installation mechanism and can be installed and updated through the Gramps Addon Manager.

DWE is installed separately from the standard *Dynamic Web Report*. Installing DWE does not replace or modify the DWR, and both can be available in Gramps at the same time.

3.1 Installing the addon

Dynamic Web Extended can be installed using the Gramps **Addon Manager**. DWE is distributed through an addon project containing versions of the addon appropriate for supported versions of Gramps.

In Gramps, open the **Addon Manager**. If the Dynamic Web Extended project is not already configured, add a DWE project under **Projects** and ensure that it is enabled. The project address must be appropriate for the version of Gramps you are using.

Once the project has been enabled, **Dynamic Web Extended** will appear in the list of available addons. Select it and install it using the Addon Manager. If an earlier version of DWE is already installed, the Addon Manager will instead indicate that an update is available and provide an **Update** button.

After installation, Dynamic Web Extended should appear under the Gramps top-level **Reports** menu, in the **Web Pages** submenu.

DWE has been tested with **Gramps 5.2** and **Gramps 6.0**. The addon supplied for each version of Gramps is built specifically for that Gramps version.

3.2 Upgrading DWE

When installing a newer version of *Dynamic Web Extended*, use the addon manager. Your genealogy data is not stored in the DWE addon directory, so upgrading the addon does not alter your family tree.

If you have manually modified files within your DWE installation, make a copy of those changes before upgrading, as addon files may be replaced during installation.

3.3 Removing DWE

DWE can be removed without affecting the standard *Dynamic Web Report* or your Gramps family-tree data. Previously generated DWE websites are also independent of the installed addon and can continue to be viewed.

4 Creating a Dynamic Web Extended website

Once installed in Gramps *Dynamic Web Extended* is run in the same way as the standard Gramps *Dynamic Web Report*.

4.1 Selecting the DWE report

In Gramps, open the family tree for which you want to create the website. Select **Reports → Web Pages → Dynamic Web Extended** from the Gramps menu.

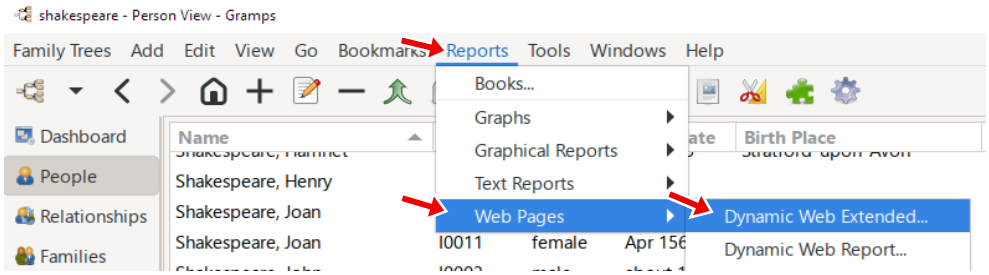


Figure 1- DWE report location

You will see the report's options pages.

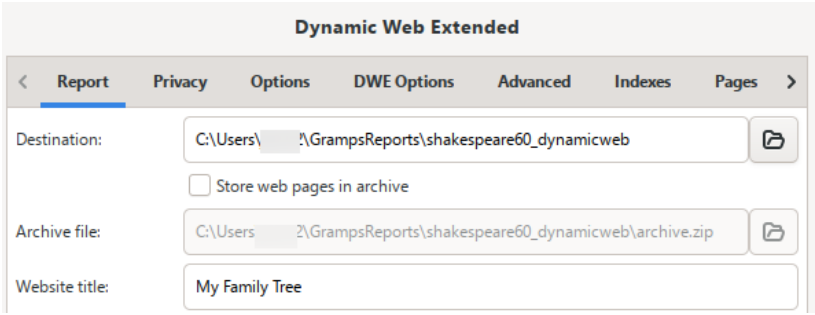


Figure 2- DWE options display

4.2 Choosing where to create the website

Choose the **Destination** directory in which DWE will create the website.

DWE generates a collection of HTML, JavaScript, data and other supporting files in the directory. These files together form the website. The generated directory and its subdirectories should always be kept together.

If you generate the report again into the same location, the website files are regenerated from the current Gramps data and the current options you have selected.

4.3 Selecting the people to include

As with the *Dynamic Web Report,* you can use a Gramps filter to determine which people are included in the generated website. The selected people, together with the other report options, determine the information that *DWE* exports for your family tree.

4.4 DWE-specific options

Many report options are inherited from the *Dynamic Web Report* and behave in the same way. *DWE* adds options for its additional features, on the **DWE Options** page.

Additional options control features such as **Events**, **Lifespans**, **Tree Viewer**, **Tracking**, **Why Me**, **Story**, **Markdown notes** and **Local Labels**. This guide explains how each option affects a feature.

The DWE-specific options for a website creator are described together in Chapter 18, DWE Options The changes to the website display from selecting those options are explained in the rest of this guide.

4.5 Generating the report

After selecting the required options, click **OK** to generate the report. Depending on the size of the family tree and the options selected, generation may take some time.

The resulting website can be copied to a web server for publication, or it can be viewed locally on a computer. Local viewing is described below.

4.6 Regenerating an existing report

A DWE report can be generated again into an existing destination directory. Files generated by DWE are updated as required. Files that have not changed may be left untouched, as with the original *Dynamic Web Report*.

Files that you have added to the destination directory yourself are not automatically removed. This also means that files created by an earlier version or configuration of DWE may remain after they are no longer required.

For this reason, when preparing a clean copy of a website for publication, it is advisable to generate the report into a new or empty directory.

4.7 Deploying the website

The website folders and files are generated by DWE into a folder on your computer. To use the website on a web server, transfer that folder and its contents to your web server.

4.8 Local viewing

A DWE website can be viewed locally on your computer or another computer. To use it on another computer, copy the whole directory in which the report was created (you selected the directory when you created the report).

To use all DWE features when viewing the report locally, it should be opened through a local web server. You don't need to create one yourself, DWE created a file to make that easy on Windows.

4.8.1 Start the website

Open the report directory and double-click **View report locally.bat**, or right-click it and select **Open**.

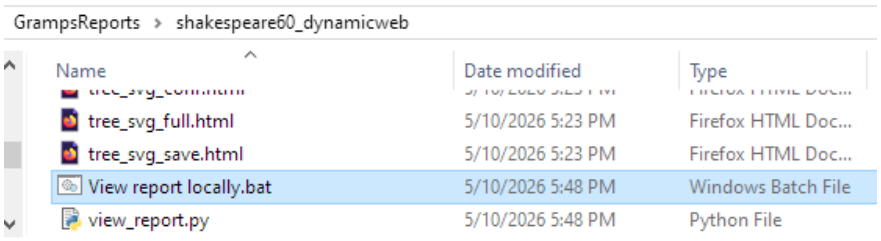


Figure 3- example batch file folder location

A small Windows command window opens and starts a local web server on your computer. The DWE home page then opens automatically in your browser. Leave the command window open while you are viewing the website.

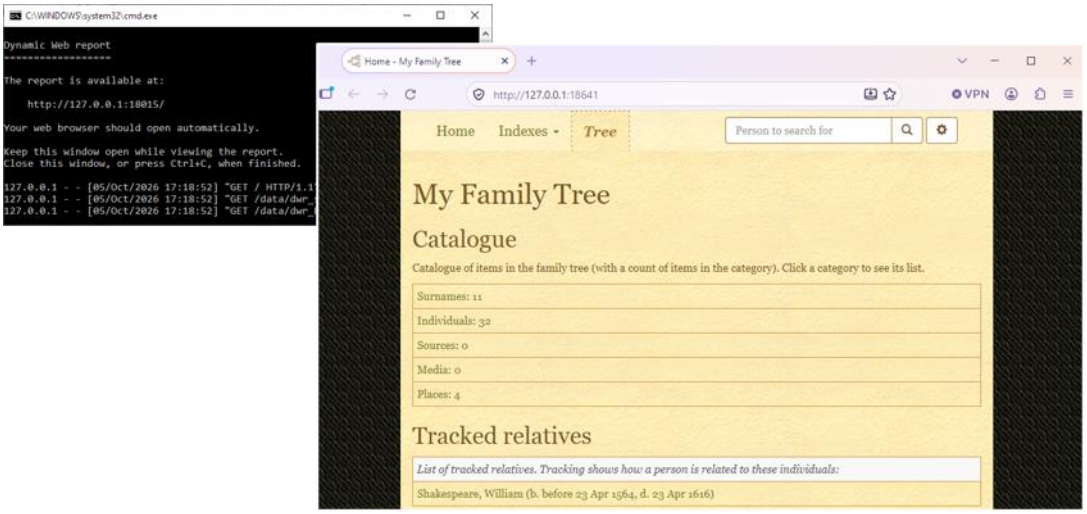


Figure 4 - Windows command window and DWE report opened in the browser

When you have finished, close the DWE browser page and then close the command window.

4.8.2 How it works

Using **View report locally.bat** starts a built-in web server and allows the **Tree Viewer** to open. It also allows **OpenStreetMap** maps to work.

OpenStreetMap requires map requests from websites to include information identifying the referring website. Opening the report through a local web server allows the browser to provide this information.

Note: The local web server runs only on your computer, and report data **is not uploaded to the internet**. Features that obtain information from external services, such as OpenStreetMap map tiles, still require an internet connection.

5 Person pages

Most of the changes in DWE can be seen in the individual's Person pages. Main features added to the page such as extended events, lifespans display and so on are described in their own chapters later in this guide.

5.1 Expandable individual details

In DWE the information at the top of an individual's page can be expanded or collapsed. By default, the details are collapsed, providing a more compact layout.

This feature stabilises the top of the page when navigating between individuals, particularly when some have more biographical information than others.

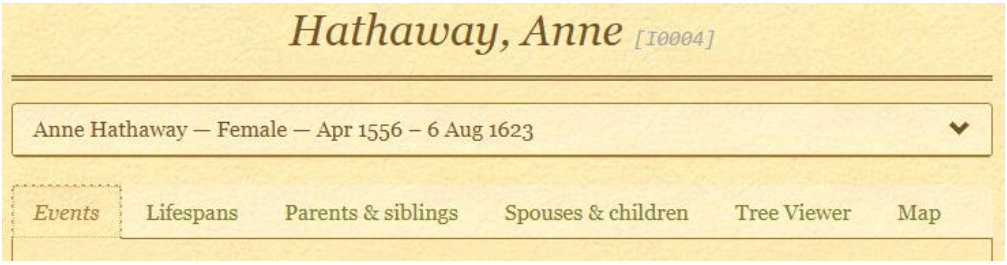


Figure 5- collapsed person details

The heading displays the individual's short name, gender and lifespan. The short name uses the format selected by the report creator in the **Report** page of the report's **Options**.

Click on the heading to expand the details:

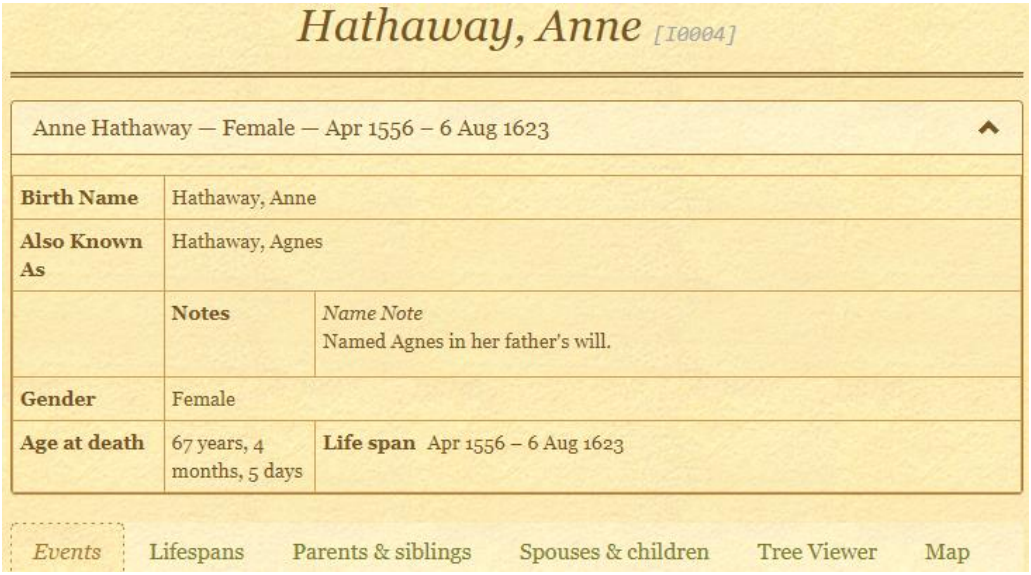


Figure 6- expanded person details

Expanded, it displays the full individual details, including additional names, biographical information and other information available for that person. Click it again to collapse the section.

DWE remembers whether the section was expanded or collapsed when you move between individuals, so you do not need to adjust it each time you open another person's page. The expandable individual-details header works independently of the tabbed-panel setting in **Configuration**.

6 Events

DWE substantially extends the **Events** display provided by the original *Dynamic Web Report*. The Events tab now combines a person's own events with important events in the lives of close family members, creating a chronological view of the person in their family context.

For example, it can show when a person's children were born, whether their parents were still alive at a particular time, or how events in the lives of siblings and spouses relate to the person's own events.

The Events table is a timeline display – earliest events are at the top, and later events fall below them, in date order:

Age	*Event*	*Date*	*Who*	*Place*	*Details*
6 yrs before birth	**Birth of spouse**	about 1531	Shakespeare, John [I0003] (husband)		*Other participants*: Shakespeare, Richard [I0015] (b. about 1490, d. before 10 Feb 1561) birth father Unknown [I0031] (b. ?) birth mother
0 mths	**Birth**	about 1537			*Other participants*: Arden, Robert [I0019] (b. ?, d. Dec 1556) birth father Webbe, Agnes [I0018] (b. ?) birth mother
19 yrs	**Death of parent**	Dec 1556	Arden, Robert [I0019] (father)		
20 yrs	**Marriage**	about 1557	Shakespeare, John [I0003]		*Other participants*: Arden, Mary [I0002] (b. about 1537, d. Sep 1608) Aide Priestly, Peter [I0052] (b. 1600, d. 1670) Celebrant

Figure 7 - Events table example

The table is like the timeline displays used by many online family-tree sites. The table lists the following events for a person:

- their individual events as usual, such as birth, residence, occupation, census, news, military service...
- birth and death events for their direct family:
  - their children
  - their spouses
  - their siblings
  - their parents
- Events in which the person has a role (called Participation events)

The events are combined into a single chronological display. Events without a date are placed at the beginning of the table.

6.1 Filtering the Events display

Checkboxes above the Events table allow visitors to control the amount of information displayed. Events can be filtered onscreen without regenerating the website.



Figure 8- Event display filters

The browser remembers the checkbox selections when navigating between person pages and when the pages are reloaded, subject to the browser's local storage settings.

6.2 Filter events for family

Events for **Children**, **Spouses**, **Siblings** and **Parents** can be shown or hidden independently. This helps to see events in the person's life in the context of changes within their family

Filtering can be useful for people with large families, where displaying every birth and death may otherwise make the timeline quite long.

- **Children** – show or hide birth and death events for the person's children.
- **Spouses** – show or hide birth and death events for the person's spouse or spouses.
- **Siblings** – show or hide birth and death events for brothers and sisters.
    - where two people share both birth parents, they are shown as brother, sister or sibling
    - where they share one birth parent, they are shown as half-brother, half-sister or half-sibling
    - If they do not share a birth parent, DWE shows them as non-birth brother, non-birth sister or non-birth sibling
    - The **Parents & siblings** tab shows the recorded parent-child relationships in more detail
- **Parents** – show or hide birth and death events for the person's parents.

6.3 Limit events to lifespan

The **Limit events to lifespan** option will restrict the events display to dated events that occurred during the person's lifetime.

- With this filter off, the table will include events before the person was born such as a parent's birth or events involving family members after the person's death.
- Events without a date are not included when the display is limited to the person's lifespan (because it is not known if they occurred within the lifespan).

6.4 Filter other information

Under **Show:** there are three independent controls:

- **Notes** — shows or hides event notes and associated descriptive details.
- **Participants** — shows or hides the other participants details.
- **Participation events** — shows or hides entire event rows representing participation in another person's event.

These selections allow a visitor to switch between a detailed display and a shorter chronological list. Turning off **Participants** hides the participant details but retains the events themselves. Turning off **Participation events** removes those events from the table.

6.5 Participants

In Gramps, several people can be associated with the same event, each with a particular role. These roles may include Primary, Witness, Clergy, Celebrant, or a user-defined role. The additional participants and their roles are displayed in the Details column.

Gramps has a default set of roles, and new roles can be assigned by the Gramps database creator. This feature is described in the Gramps wiki.

6.6 Who and Details columns

6.6.1 Who

The **Who** column identifies the principal people associated with an event, excluding the person whose page is shown.

- For a person's own individual events, such as birth, death or residence, the Who column is normally empty because it is their page.
- For family events, such as marriage, the Who column identifies the other partner. For events involving relatives, it identifies the relevant family member.
- For Participation events, the Who column identifies the principal people associated with the event.

6.6.2 Details

The **Details** column identifies other participants and their roles, such as witnesses, clergy or celebrants.

- A person who has more than one role is listed once, with their additional roles grouped together (for example, Best man, Witness).
- Additional roles of a primary participant are displayed in Details.

6.7 Participation Events

An event in which a person does not have a Primary or Family role is displayed as a **Participation event**. Its event type appears on the first line, with the person's role or roles on a second line, for example Marriage followed by (:Best man).

Suppose Joan Shakespeare died in November 1646, with Charles Cadbury recorded as a witness and Peter Priestly as the officiating clergy.

The event appears on their respective person pages as follows. The entries all refer to the same Gramps death event; the (:Participation) label makes it clear that the event did not happen to Charles or Peter, but that they were involved in it.

- **Joan Shakespeare's page:** Death event

Age	*Event*	*Date*	*Who*	*Place*	*Details*
77 yrs	**Death**	Nov 1646			*Other participants*: Cadbury, Charles [I0051] (b. ?) Witness Priestly, Peter [I0052] (b. 1600, d. 1670) Clergy

- **Charles Cadbury's page:** Death (:Witness)

Age	*Event*	*Date*	*Who*	*Place*	*Details*
	Death (:Witness)	Nov 1646	Shakespeare, Joan [I0011]		*Other participants*: Cadbury, Charles [I0051] (b. ?) Witness Priestly, Peter [I0052] (b. 1600, d. 1670) Clergy

- **Peter Priestly's page:** Death (:Clergy)

Age	*Event*	*Date*	*Who*	*Place*	*Details*
46 yrs	**Death (:Clergy)**	Nov 1646	Shakespeare, Joan [I0011]		*Other participants*: Cadbury, Charles [I0051] (b. ?) Witness Priestly, Peter [I0052] (b. 1600, d. 1670) Clergy

The Who column identifies Joan as the principal person on Charles's and Peter's pages.

6.8 Marriage events

Marriage is recorded in Gramps as a Family event rather than as a Person event. The Events table displays marriage and other Family events in the person's timeline. This means it is not necessary to add the same marriage event separately to a person's individual events in Gramps simply to make it appear in their Events display.

Other people associated with the marriage, such as witnesses or celebrants, will have the marriage in their own Events tables as a Participation event.

6.9 Maps of Events

Maps are available on Person, Family and Place pages when geographical coordinates have been recorded.

- The **Person Map** shows locations associated with the person's own events, including marriages and events in which they explicitly participated. Events belonging solely to relatives are excluded.
- The **Family Map** shows locations associated with family events and individual events of the parents and children in that family.
- The **Place Map** identifies the geographical location of the selected place. It does not show events.

Like DWR, when several events occur at the same coordinates, they are combined into one map marker. Selecting the marker displays the associated people, families and events.

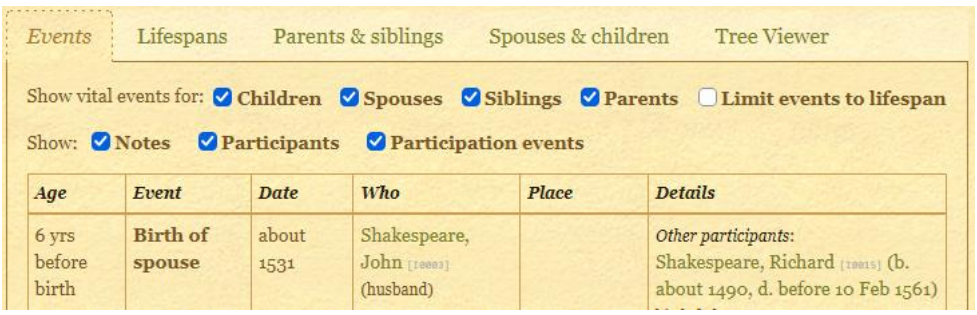
6.10 About…

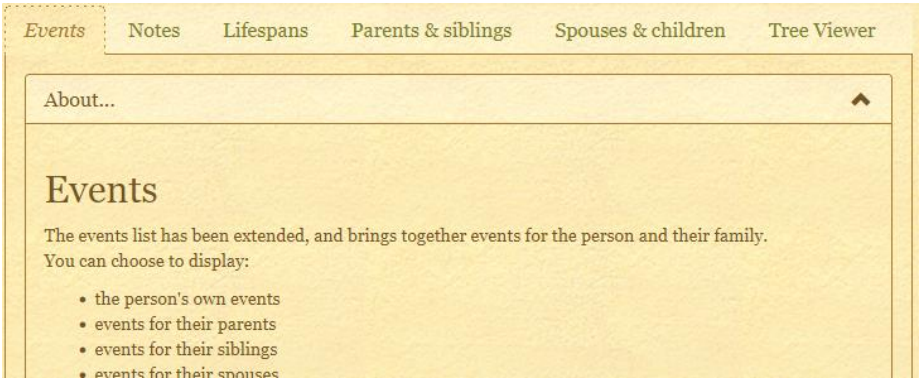
Because DWE adds several controls and family events to the original Events display, you can provide an explanation for people using the website.

When creating the report, an optional Gramps note can be selected using **Events note** on the **DWE Options** page. If selected, it is displayed in a collapsible **About…** section on the Events page. If no note is selected, the About section is not displayed.

The note can explain which family events are included, how the display controls work, or anything else that may help people using the website. See About... notes in Chapter **18 DWE Options** for information on selecting and using these notes.

About note appearance

<table>
<tr><td>No note selected</td><td> 
</td></tr>
<tr><td>Selected
(About is collapsed)</td><td> 
</td></tr>
</table>

Selected (About is expanded)	

The text is displayed from the note, and the note can say whatever you want, in whatever language you need. An example of note content is shown at the end of this document.

6.11 Remembering display settings

The settings of the check boxes are saved and remembered as you open or reload a page. So, if Siblings is turned off, it is still turned off for the next person you open. Also, if you close a tab, or close the browser, the settings are remembered for next time.

If you already have a couple of person pages open, then changing the settings in one page won't update the settings of the other opened page, unless/until you reload that other page.

This saved tick behaviour depends on your browser's local storage settings.

7 Lifespans

The **Lifespans** tab provides a graphical view of a person's life in relation to the lives of their immediate family. It makes it easy to see how the person's lifespan overlaps those of their parents, siblings, spouses and children.

The display consists of horizontal lifespan bars arranged against a common time scale. The person whose page is being viewed is the **subject** of the display.

If you hover over a person's lifespan bar (or either side of it) their birth and death dates and places are displayed.

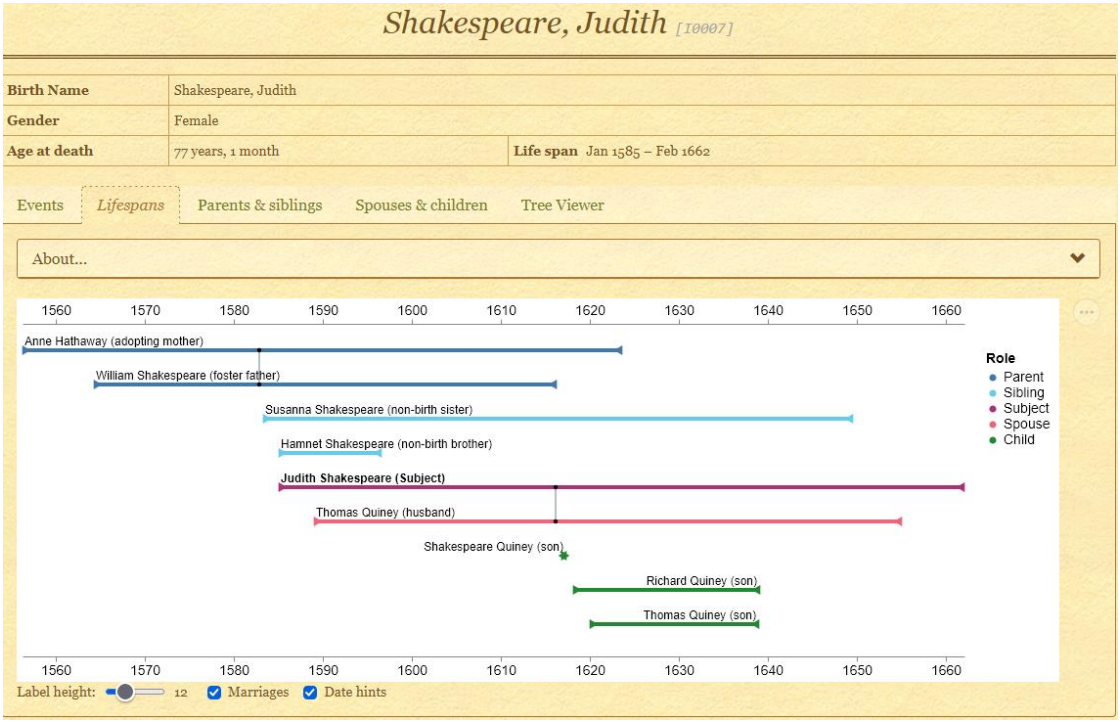


Figure 9 - Lifespans chart

7.1 People shown

The Lifespans display groups people according to their relationship to the subject:

- **Parents**
- **Siblings**, including the subject
- **Spouses**
- **Children**

The subject is shown among their siblings, making it easier to see the order and overlap of the children's lives within the family.

A person can belong to more than one family or have more than one set of parents. DWE uses the parent-child relationships recorded in Gramps when finding the people to include and when describing their relationship to the subject. These can include birth, adoptive, foster, step and other parent-child relationships explicitly recorded in Gramps.

A person who occurs more than once as a child of the subject is shown only once in the Lifespans display.

Relatives are grouped by their relationship to the subject rather than by individual family. The **Parents & siblings** and **Spouses & children** sections on a person's page can be examined when the individual family structures and recorded relationships need to be viewed in more detail.

7.2 Lifespan bars and markers

Where both birth and death dates are available, a horizontal bar shows the person's lifespan between the two dates. Birth and death are also indicated by markers on the timescale.

If only a birth date or only a death date is available, a marker is shown for the known date, but no lifespan bar can be drawn. When birth and death are very close together, the markers are displayed in a form that makes them easier to distinguish.

Each person is identified by name and by their relationship to the subject. The different family roles are also distinguished by colour. The colours are shown in a Legend on the right.

7.3 Dates and ordering

Parents, siblings and children are ordered by birth date where available. Spouses are ordered by marriage date where available. People for whom the relevant date is unknown are placed after those with known dates and are still included in the display, rather than being discarded. This allows the chart to remain useful even when some dates in the family tree are incomplete.

7.4 Timescale

The Lifespans timescale extends from the earliest to the latest date that can be plotted for the people included in the chart. Birth, death and marriage dates affect the extent of the timescale.

To make it easier to compare a person's dates with the timescale and with other lifespans, hover over the person's name. Vertical **date hints** extending towards the timescale are displayed and change as you move from person to person.

The date hints can be shown or hidden using the **Date hints** checkbox. The setting is remembered by the browser.

7.5 Marriage events

Marriage events are shown on the Lifespans display so that marriages can be seen in the same time context as births, deaths and other family members' lifespans. Hover over a marriage marker to display its date and place. The marriage marker is a vertical line joining the two participants, placed at the appropriate date if known.

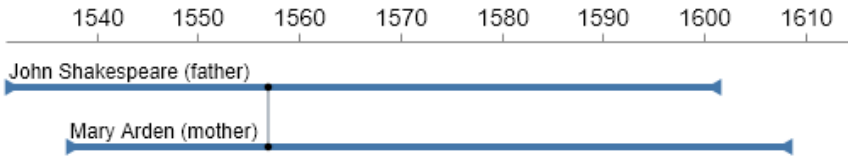


Figure 10- marriage marker

Marriage events can be shown or hidden using the **Marriages** checkbox. The setting is remembered by the browser.

Undated marriages

When a marriage is recorded in Gramps but has no usable date, it is displayed in a separate area to the left of the dated timeline. If several undated marriages are present, their connecting lines are spaced horizontally to distinguish them. If there are no undated marriages, the separate area is omitted.

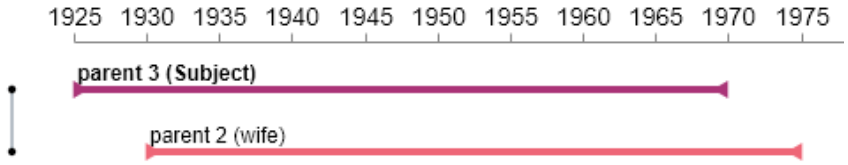


Figure 11- display of an undated marriage

7.6 Label height

Depending on the size of your screen and the number of people shown in the chart, you may want to adjust the label size to make the chart easier to read.

The **Label height** slider changes the font size of the person labels. The selected label height is remembered by the browser and is used for other Lifespans charts you view.

Changing the label height does not change the vertical spacing between people.

7.7 Moving to another person

Each person's name in the Lifespans display is a link. Select a name to move to that person's page. Their page opens with the last view you used for that page. You can select their **Lifespans** tab to see the chart for that person and their family.

7.8 About…

The website creator can provide an optional note containing information about the Lifespans display. When a Lifespans note has been selected in the DWE Options, it is displayed under **About...** If no note has been selected, the About section is not shown.

7.9 Saving the chart

The Lifespans chart can be saved as an image. Select the three-dot menu at the upper right of the chart and choose **Save as SVG** or **Save as PNG**. The files are named `lifespans.svg` or `lifespans.png` and are normally saved in the browser's Downloads folder.

SVG is particularly useful if the chart is to be enlarged, printed or edited later, because it is a vector format and can be resized without losing quality. PNG is convenient for ordinary image use.

8 Tree Viewer

DWE provides an interactive graphical **Tree Viewer** for exploring relationships within the family tree. It uses a version of the **Topola Genealogy Viewer** that is bundled with DWE.

Topola is the tool already used by Gramps to generate the interactive family tree available from the main Gramps menu under **Tools → Analysis and Exploration → Interactive Family Tree**. Tree Viewer makes that view available to a DWE website visitor.

8.1 Tree and Tree Viewer

DWE provides two different interactive tree displays:

- **Tree** is the family-tree display inherited from the original *Dynamic Web Report*. It provides extensive configuration options for controlling the content and appearance of the tree. Tree can be opened from the top-level menu bar on any report page.
- **Tree Viewer** is the additional DWE display based on Topola. It provides a different way of exploring the family tree, allowing you to navigate through the chart and select people directly from the graphical display. Tree Viewer can be opened by selecting the Tree Viewer tab on a person's page.

The two displays are independent. **Tree** is the familiar *Dynamic Web Report* tree, while **Tree Viewer** provides another way of navigating and exploring the genealogy data.

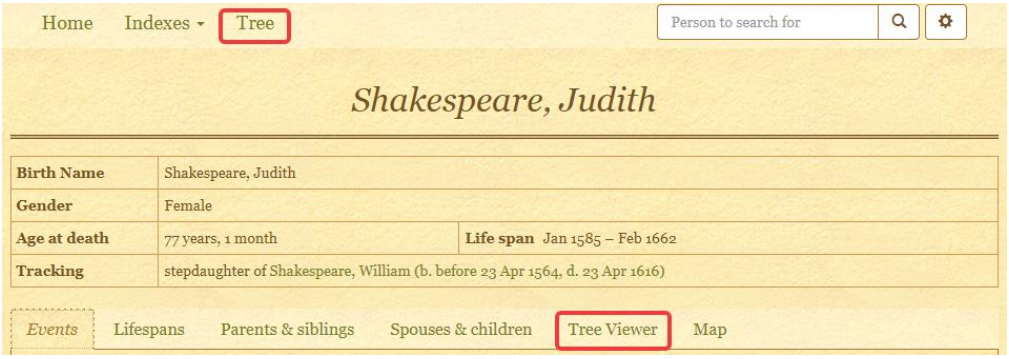


Figure 12 Person page showing Tree and Tree Viewer controls

8.2 Setup

Tree Viewer is set up by the DWE website creator. When creating the DWE report, the creator will select the option to include Tree Viewer on the website. See Include Tree Viewer in Chapter 18 of this guide.

8.3 Using the Tree Viewer

When Tree Viewer is opened, the tree is initially centred on the person whose DWE page you are viewing. That person is highlighted within the tree. A side panel on the right shows details about that person.

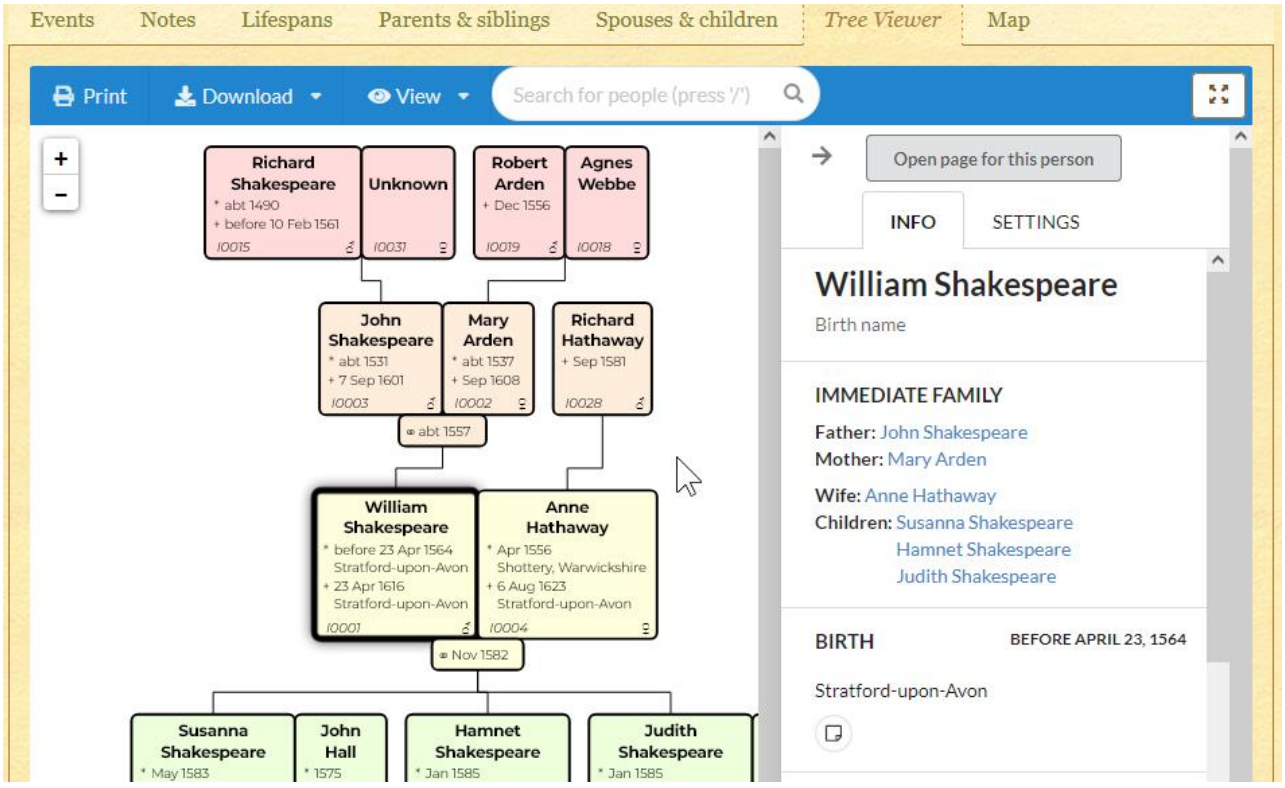


Figure 13- Tree Viewer

The Tree Viewer display and controls are very similar to the Gramps **Interactive Family Tree** display. You can pan and zoom the display and select people in the tree to explore their family relationships. You can collapse the side panel by selecting the → arrow icon at the top of the side panel. Click the icon again to redisplay the panel.

If you select another person in the display, the tree is reorganised and the new person is highlighted. You can also click on a name in the side panel to move to a new person.

As you move around the tree the side panel details change to match the person highlighted in the tree, but you are still within the original person's DWE page. The side panel contains an **Open page for this person** button. Selecting the button opens the highlighted person's DWE page.

The tree is initially displayed within the Tree Viewer section. To give the tree more space, select the **Maximize** icon  at the top right of the Tree Viewer. The viewer expands to fill the browser window, and the button changes to **Restore**. Select **Restore** to return to the normal display.

8.4 Tree Viewer controls

The toolbar across the top of the Tree Viewer provides the standard Topola controls:

- **Print** — opens the browser's print facility for printing the displayed family tree.
- **Download** — saves the displayed tree as a PDF, PNG or SVG file. The files are named `DWEtree.pdf`, `DWEtree.png` or `DWEtree.svg`, and are normally saved in the browser's Downloads folder.
- **View** — selects the way the family tree is displayed. The available views are described below.
- **Search** — searches for a person in the tree. Partial names can be used. Selecting a person from the search results changes the focus of the tree to that person.
- **+** and **−** — zoom in and out.

You can also drag the tree to move around the display.

8.5 Choosing the tree view

Select **View** to change the way the family tree is displayed. Four views are available:

- **Hourglass chart** — displays ancestors and descendants around the selected person. This is the normal Tree Viewer display.
- **All relatives** — displays a wider range of people related to the selected person.
- **Donatso family chart** — provides an alternative graphical arrangement of the family.
- **Family tree (experimental)** — displays people graphically on a drawing of a tree. This view is most practical when displaying a relatively small number of people.

The selected view is not remembered when you leave the Tree Viewer. When you open the Tree Viewer again, it starts with the **Hourglass chart**.

8.6 Tree Viewer data

When Tree Viewer is enabled Tree Viewer and the data it requires are automatically created. A website visitor does not need to install Topola, visit the Topola website, or provide a separate GEDCOM file

- DWE generates a GEDCOM file containing the genealogical data required by the viewer.
- DWE removes the Gramps researcher copyright information from the GEDCOM.
- That file and a bundled Topola viewer with its supporting files are placed in the generated website.

8.7 About...

The website creator can provide an optional note containing information about the Tree Viewer. When a **Tree Viewer note** has been selected in **DWE Options**, it is displayed under **About...** in the Tree Viewer section. If no note has been selected, the About section is not shown.

9 Tracking

Tracking shows how people in the DWE website are related to one or more selected people. It can be useful when exploring a large family tree, because it provides a quick indication of how the person you are viewing connects to people whose relationship to you is already familiar.

For example, you might track yourself and your partner, or several people representing different branches of the family. In my family tree I track four people — my two parents and my partner's two parents. When exploring my partner's side of the family, this makes it easy to see how people are related to her parents.

DWE uses a Gramps tag to identify the people to be tracked. You can track as many people as you wish, or not use Tracking at all.

9.1 Set up

Tracking is set up by the DWE website creator. When creating the DWE report, the creator will select a tag as the **Tracking tag** value. See Tracking tag in Chapter 18 of this guide.

9.2 Result

9.2.1 Tracked relatives listed on the home page

When Tracking is enabled, the people in the report who have the Tracking tag set for them are listed as **Tracked relatives** on the DWE home page. If no Tracking tag has been selected, the section is not displayed.

If a Tracking tag has been selected but no people in the report have that tag, the **Tracked relatives** section displays a message to alert the website visitor.

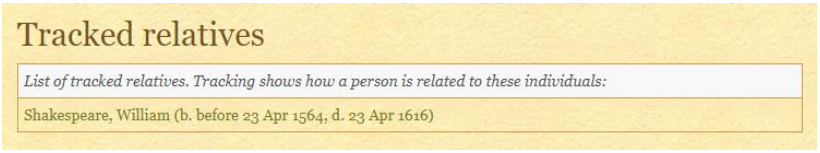


Figure 14- no tracked person found

9.2.2 Tracking entry on a person's page

On a person page, the **Tracking** entry shows their connection to each tracked person. If the person is related to more than one tracked person, a separate entry is shown for each tracked relative.

The tracked person's name is a link. Select their name to open their DWE page.

A tracked person is not shown as being related to themselves, but their relationships to other tracked people can still be shown.

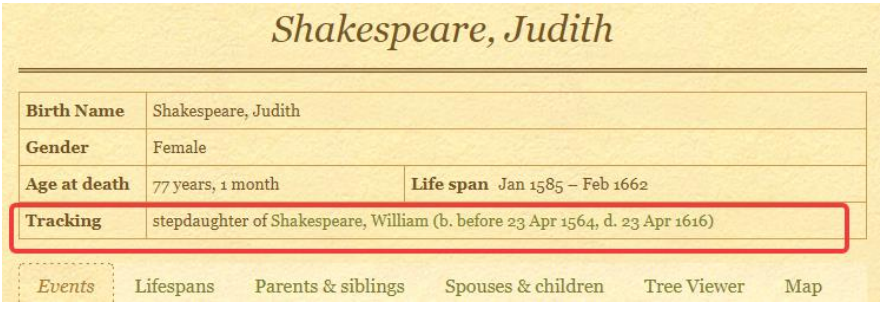


Figure 15- Tracking of William Shakespeare shows Judith is a stepdaughter

10 Why Me

People can be included in a family tree for many reasons. They may be relatives, but may also be other people like friends and neighbours who are important to the family history, though not related.

Why Me provides a way for the website to explain why a particular person is included in the tree. The explanation is stored as a Gramps note attached to that person and is displayed with the person's details on their DWE person page.

Why Me is independent of Tracking. Tracking uses the Gramps relationship calculator to show a person's genealogical relationship to selected tracked people. Why Me is information supplied by the creator of the family tree. A person can therefore have a Tracking relationship, a Why Me note, both, or neither.

10.1 Set up

Why Me is set up by the DWE website creator. When creating the DWE report, the creator will select a Gramps note type as the **Why Me note** type. See Why me note type in Chapter 18 of this guide.

10.2 Result

If a person has a note of the selected type, its contents are displayed as **Why me** in the person's details.

Birth Name	Sadler, Hamnet		
Gender	Male		
Age at death	about 62 years	**Life span**	about 1562 – 1624
Why me	A local baker and close childhood friend. Shakespeare named his twins, Hamnet and Judith, after him and his wife.		

Figure 16- Why me note display on person page

The Why Me note is a Gramps note and will also appear within the person's Notes tab. The Notes tab will be displayed if the website creator has selected the **Export Notes option** on the **Privacy** page of the report's options. That option is inherited from the *Dynamic Web Report*.

The **Why me** label can be changed using DWE's Local Labels feature, described in Chapter 13. For example, a website creator might replace the short **Why me** label with a longer explanation appropriate to the website.

I use Why Me notes for many people in my own tree, including people for whom Tracking relationships are also available. The two features provide different information: **Tracking describes a calculated relationship, while Why Me explains why I have included the person in the tree**.

11 Story

The **Story** feature provides a place for a short introduction or summary, displayed with the person's details near the top of their page.

11.1 Set up

Story is set up by the DWE website creator. When creating the DWE report, the creator will select a Gramps note type as the **Story note** type. See Story note type in Chapter 18 of this guide.

11.2 Result

If a person has a note of the selected type, its contents are displayed as **Story** in the person's details. If the person does not have a Story note, the Story entry is not displayed.

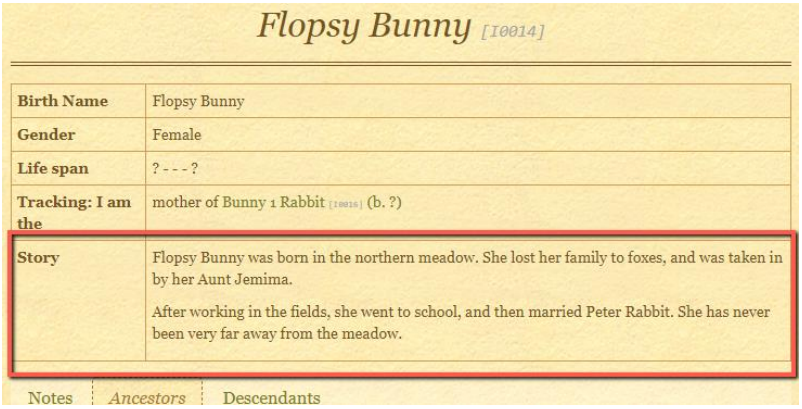


Figure 17- Story note display

Story notes are ordinary Gramps notes and will also appear with the person's notes in the **Notes** tab.

I keep Story notes fairly short — only a few sentences. Their purpose is to provide a quick introduction to the person rather than a complete biography. A long Story requires the visitor to scroll past it before reaching the other information on the person's page.

For longer stories, I prefer to create a separate document. For example, I use Microsoft Word's document features, including headings and images, save the finished document as a PDF, and add the PDF to Gramps as media. It can then be linked to the person in the same way as other media.

12 Clarifying multiple families

The original *Dynamic Web Report* can show people who belong to more than one family and can display the parent-child relationships recorded in Gramps. However, I found that some less usual family arrangements could be difficult to follow, particularly when several parent families and their children were displayed together.

DWE retains this information but changes parts of its presentation to make the family connections clearer.

12.1 Parents & siblings

The **Parents & siblings** tab, called **Ancestors** in the original *Dynamic Web Report*, shows the families in which the person is recorded as a child. The display nests children in their appropriate parent family, making it easier to see which people belong to each family. Collapsing a family also collapses its children.

Family type is displayed (such as married or unknown) and parent-child relationships are named. DWE uses the relationships recorded in Gramps and does not try to infer a step relationship from the surrounding family structure.

12.2 Children from other families of the parents

By default, DWE also shows children from other families of the person's parents. This makes half-sibling relationships and other family connections easier to understand.

For example, if a person's mother has children in another family, that family can also be shown in **Parents & siblings**, even though the person whose page is being viewed is not a child in that family. DWE makes this clear with the message:

> **The person on the page is not a child in this family**
> (This family is another marriage/union of a parent)

The report option **Include children from other families of the parents** controls this behaviour. It is selected by default in DWE. The option appears on the report's **Advanced** options page. Clear the checkbox if you want **Parents & siblings** to show only the families in which the person is recorded as a child.

This option affects only the **Parents & siblings** display. Other parts of DWE continue to use the relationships recorded in the family tree. For example, **Tracking** may identify someone as a half-sibling, and **Events** may include events for that half-sibling, even when the other family has been hidden by clearing this option.

12.3 Families without recorded parents

Sometimes you may have discovered that two people are siblings without knowing the identity of either parent. Gramps allows the children to be placed together in a family without adding a father or mother.

DWE can display families without parents in **Parents & siblings**, allowing the known sibling relationship to be shown without requiring an artificial unknown parent to be created.

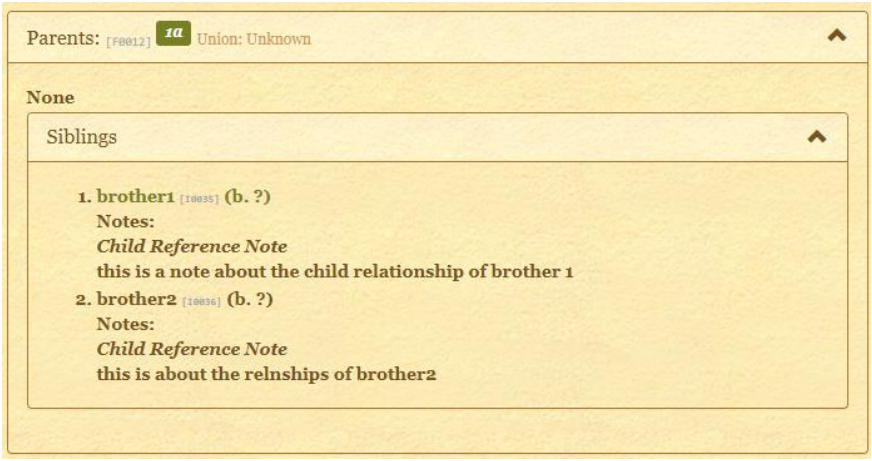


Figure 18- family without parents

12.4 Spouses & children

The **Spouses & children** tab, called **Descendants** in the original *Dynamic Web Repor*t, shows the person's families as a spouse or partner and the children in those families. DWE adds the marriage or family relationship type to the family display.

13 Local Labels

DWE allows the website creator to replace labels used in the generated report without editing the program code. This can be useful if you prefer different terminology or want labels that are more appropriate for a particular family history website.

13.1 Setup

Local Labels is set up by the DWE website creator. When creating the DWE report, the creator will select a specific Gramps note as the **Local labels note**. See Local labels note in Chapter 18 of this guide.

Local Labels are chosen by the person creating the report. Website visitors cannot change the labels themselves.

14 About page for website

The DWE top level menu has an About page that displays version details, for diagnostic purposes. It also includes a link to a built-in visitor guide. The guide can be edited. See Chapter 15 Visitor Guide.

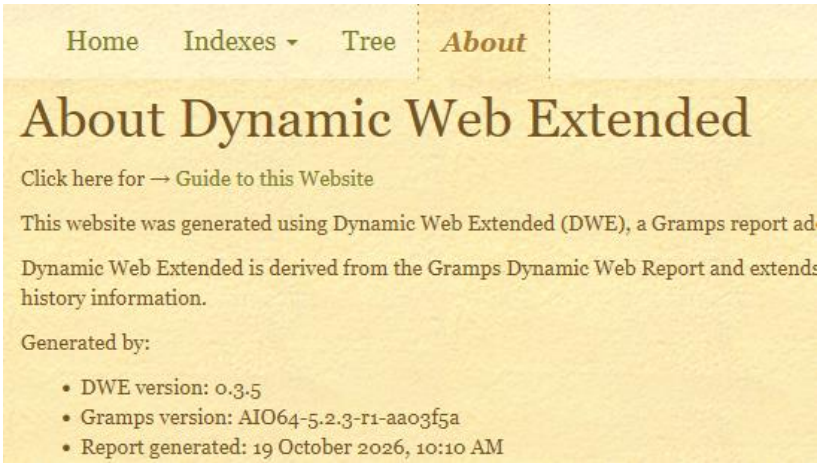


Figure 19- example About page

15 Visitor Guide

Dynamic Web Extended (DWE) provides a facility for adding an editable webpage to a previously generated website.

The **Visitor Guide** is a webpage whose content is stored in a separate file distributed with the website. The website creator, or anyone with access to the generated website files, can edit its contents without regenerating the report.

Visitors can open the Visitor Guide using a hyperlink on the website's **About** page.

Note that DWE retains the existing Custom Page facility from the Gramps *Dynamic Web Report* (DWR). Custom Page allows the website creator to add one or more pages to their website menus using content from Gramps notes. The Custom Page facility is described in the DWR documentation on the Gramps wiki.

15.1 Key difference

DWE now has two complementary ways of adding formatted content to a website:

- **Custom pages** – sourced from Gramps notes, edit the notes and regenerate the report to see changes;
- **Visitor guide** – source is an external file, edit the file and refresh the browser to see changes

A more detailed comparison is shown below:

Custom pages	Visitor guide
From a Gramps note	From a Markdown file
Are included in the website's navigation menus	Accessed through a hyperlink on the About menu page
Content is maintained inside Gramps	Content is maintained in a file within the website folders
Content can include plain text, HTML, Gramps text styling and Markdown	Content styling can include plain text and Markdown formatting

Changes require editing the Gramps notes in Gramps and regenerating the report	Changes require only editing and saving the accessible Markdown file
Useful for genealogy-related content	Useful for website instructions and visitor information

15.2 Link on About page

The website About page includes a link to the Visitor Guide.



Figure 20- link to Visitor Guide

15.3 Setup

DWE automatically creates a Visitor Guide webpage when a report is first generated.

15.4 Editing

The Visitor Guide is stored in an editable Markdown file (visitor_guide.md) in the website's main directory. Its contents are converted to HTML whenever the Visitor Guide page is opened. The file may contain ordinary plain text or Markdown formatting, including headings, lists, hyperlinks and tables.

When a website is first generated, DWE copies its internal default text (visitor_guide_default.md) into the website directory as visitor_guide.md.

The website creator can then edit visitor_guide.md to customise the guide. Whenever DWE regenerates the website into the same directory, it preserves the current file, and does not replace it with the default.

15.5 Publishing

If the website is published to a remote server changes made directly to the remote server's copy are not automatically reflected back into the DWE target directory. Care should therefore be taken when uploading a regenerated website to avoid overwriting edits made on the server. If the website is copied to another computer, the same caveat applies.

16 Website Themes

The website is created using the theme selected by the report creator when generating the report. See section 18.11 Website themes.

A site visitor can change the appearance of the website at any time by selecting another theme on the DWE **Configuration** page. Open the **Configuration** page by selecting the gear icon at the top right of a site page.

Two appearance options are available.

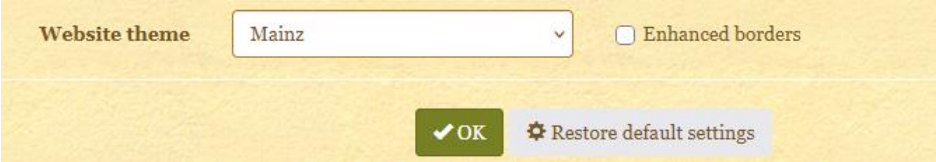


Figure 21- website themes and enhanced borders

16.1 Selecting a theme

Seven themes are listed:

Default, Mainz, Cerulean, Flatly, Journal, Readable and United.

To change the theme, select the preferred theme from the **Website theme** dropdown list, and click **OK**. The website will reload using the selected theme.

The selected theme is remembered by the browser and will normally be used on subsequent visits to the website.

The new themes are from Bootswatch. See the Bootswatch web site for more information on each theme.

Note: If you just want to change any setting on the Configuration page then change it and click the **OK** button. **The Restore default settings**, followed by **OK** restores all the Configuration settings to their defaults.

16.2 Enhanced borders

The **Enhanced borders** option on the **Configuration** page makes the boundaries around sections, tables and tabs more prominent. This can improve readability, particularly with themes that use subtle borders.

To enable this option, select the **Enhanced borders** checkbox and click **OK**. The website will reload with stronger borders. To turn off enhanced borders deselect **Enhanced borders** (no tick showing) followed by **OK**.

The setting is independent of the selected website theme and is remembered by the browser in the same way as the theme selection.

16.3 Local server limitation

Note for visitors running a local web server: If the website is opened using a local server that selects a different port each time it starts, the browser may not remember the previously selected theme and the Enhaced borders setting.

The `View report locally.bat` feature (see section 4.8 Local viewing) is set up to use a different port each time it starts.

In this case, the website will open using the report creator's default theme, and the visitor will need to select their preferred theme again.

17 Other enhancements

DWE includes several smaller changes in addition to the features described elsewhere in this guide.

- **Home page categories** — The categories on the DWE home page are presented in a table to make the list easier to read.

- **Table page sizes** — DWE adds **20** to the choices available in the **Show Entries** selector on table pages, providing an intermediate choice between 10 and 50 entries.
- **Person Index sorting** — The Person Index can be displayed as either a table or a list. In table view, the columns can be sorted using the sort controls in the column headings. In list view, sorting is controlled by the **Sort by** selector. DWE adds choices for sorting by birth or death date with either the earliest or latest dates first.
- **Dates** — Hyphens within dates are non-breaking, so a date won't split across two lines.
- **Life span** — The original DWR displays a person's **Age at death** in their person-page details. DWE also displays their **Life span**, showing the period from birth to death. When both are available, Age at death and Life span are displayed together on the same row.
- **Remembering the tabbed-panel setting –** The Configuration page includes an option to display information using either tabbed panels or separate linear sections. DWE now remembers the visitor's selection when navigating between individuals and other pages, and when returning to the website in a later browser session. The selection remains in effect until it is changed in Configuration.
- **Searching for people** – The name search includes nicknames recorded in both primary and alternative names. You can search using a nickname alone or combine it with a surname.

18 DWE Options

Dynamic Web Extended adds a **DWE Options** page to the report options. This page brings together the settings for features added by DWE. Options inherited unchanged from the *Dynamic Web Report* remain on their normal option pages and are not described in detail here.

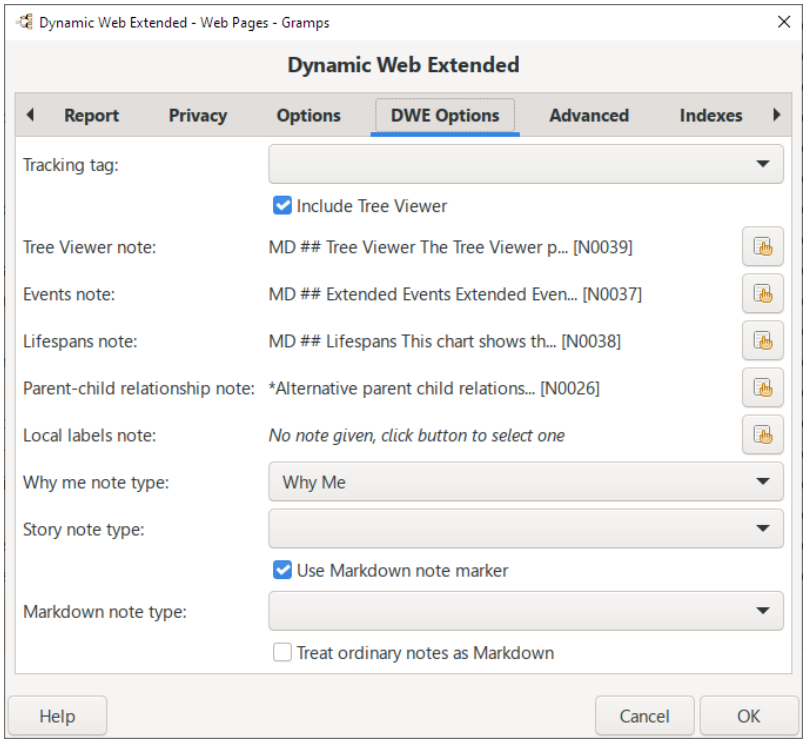


Figure 22- DWE Options

18.1 Option defaults and saved settings

When creating a DWE report for a family tree for the first time, DWE provides initial values for its options. Gramps saves the options. If you change them, they are reused the next time you create a DWE report for that family tree. Each family tree therefore has its own saved report settings.

**Notes used only to configure DWE can conveniently be attached to a specially created dummy person in the tree, such as a person named 'Gramps'. This prevents them from appearing as unused objects when Gramps database repair or checking tools are used. Examples include the Tree Viewer, Events, Lifespans, Parent-child relationship and Local labels notes.*

The initial settings for a new family tree are:

DWE option	Initial setting
Tracking tag	Empty
Include Tree Viewer	Selected
Tree Viewer note	Empty
Events note	Empty
Lifespans note	Empty
Parent-child relationship note	Empty
Local labels note	Empty
Why Me note type	Empty
Story note type	Empty
Use Markdown note marker	Selected
Markdown note type	Empty
Treat ordinary notes as Markdown	Not selected

18.2 Tracking tag

The **Tracking tag** identifies the tag to be used when DWE calculates Tracking relationships. Tracking shows the connection of each person in the report to the tracked individual or individuals.

DWE tracking uses a Gramps tag. Instructions on how to create and manage Gramps tags are provided in the Gramps Wiki Manual.

See Chapter 9 Tracking for a description of the information displayed by this feature.

18.2.1 Create a Gramps tag

Select **Edit → Tag → New Tag** from the Gramps top menu. Create or choose a Gramps tag to identify the people you want to track. The tag can have any name; for example, you might call it 'Track me'. You can use an existing tag.

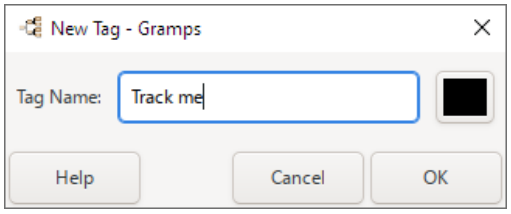


Figure 23- create a tracking tag

18.2.2 Set the Tracking tag value

Select an existing tag in the **Tracking tag** option.

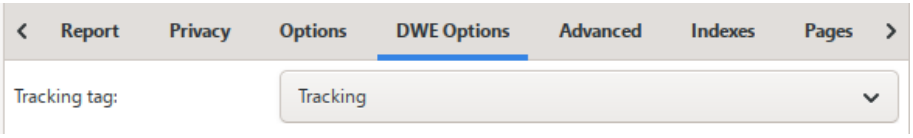


Figure 24 - select the tag to be used

Tracking is optional. To turn it off, select the blank entry for **Tracking tag** before generating the report again. If the option is left empty, **Tracking** entries are not generated.

18.2.3 Assign the Tracking tag

In the Gramps person editor, for each person you want to be tracked, select the **Tag** button and assign the 'Track me' tag to that person.

For example, you might assign that tag to William Shakespeare.

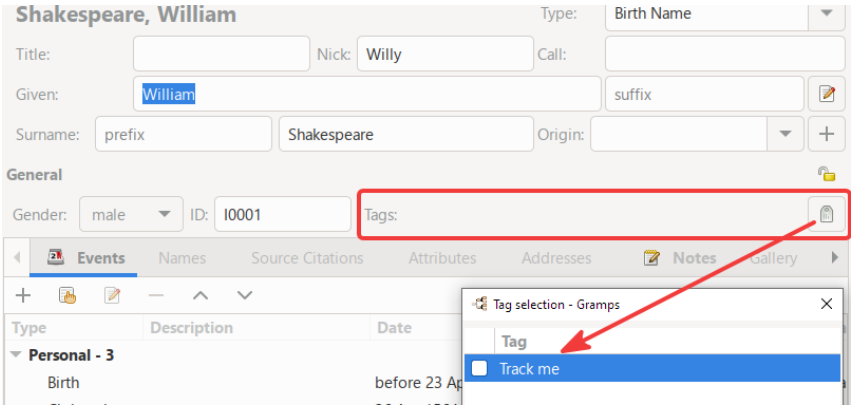


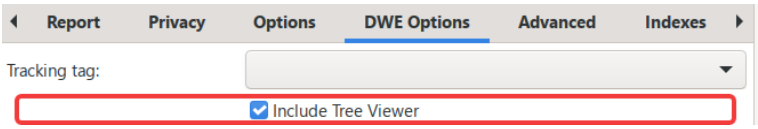
Figure 25- apply tagging on the Person Editor page

The Tracking entry on a person's page can then show their relationship to William Shakespeare, such as son, sister or wife. The web page for Susanna Shakespeare would show this Tracking entry:



18.3 Include Tree Viewer

Select **Include Tree Viewer** to include an interactive **Tree Viewer** in the generated website.



When this option is selected, DWE adds a Tree Viewer tab to the website's person pages and automatically generates the GEDCOM file required by this viewer.

Clear the checkbox if the Tree Viewer is not required.

Warning — Tree Viewer performance with large family trees

The Tree Viewer uses the Topola genealogy viewer, which may have difficulty displaying large or complex family trees. In some cases, the tree may fail to render, and the browser may become unresponsive.

During testing, a family tree containing approximately 1,100 people displayed correctly, while a test tree containing approximately 4,600 people failed to render and caused the browser to become unresponsive.

The same behaviour was observed in Gramps' **Tools → Analysis and Exploration → Interactive Family Tree**, which also uses Topola.

These observations do not establish a fixed maximum tree size. Performance may also depend on the structure and complexity of the family tree.

Recommendation: Before enabling Tree Viewer in DWE, test your family tree using Gramps' **Interactive Family Tree** tool. If it does not display correctly there, it may also fail in DWE.

See Tree Viewer in Chapter 8 for more information about using this feature.

18.4 About... notes

The next three options — **Tree Viewer note**, **Events note** and **Lifespans note** — all use the same DWE **About note** concept.

An About note is an optional Gramps note containing information that the website creator wants to provide to visitors for each of these three features. It might explain how to use the feature and what information can be seen. Example **About notes** are provided in the Appendix.

The features themselves do not require an About note. Leaving the corresponding option empty simply means that there is no creator-supplied About note, and an About box will not be displayed.

18.4.1 Tree Viewer note

Select an optional Gramps note to provide the About note for the **Tree Viewer** display.

18.4.2 Events note

Select an optional Gramps note to provide the About note for the **Events** display.

18.4.3 Lifespans note

Select an optional Gramps note to provide the About note for the **Lifespans** display.

18.5 Parent-child relationship note

Gramps records the relationship of a child to each parent in a family, for example Birth or Adopted. DWE extends that to name the relationship in both directions — from a parent to a child and from a child to a parent. The extended relationship phrases are used for wording in the Events tables and in the Lifespans charts.

DWE contains default phrases for the standard Gramps relationship types. The optional **Parent-child relationship note** allows the website creator to override a default and use different wording for the relationship. If you don't need to change the defaults, you don't need a note.

Each relationship is placed on a separate line of the note. A line has 5 fields separated by '>':

```
Gramps relationship > parent phrase > child phrase > kinship position > optional comment
```

The first field is the Gramps relationship name. The second is the phrase used for the parent relationship and the third is the phrase used for the child relationship.

The fourth field controls whether DWE adds the appropriate **kinship term**, such as *father*, *mother*, *parent*, *son*, *daughter* or *child*:

Value	Meaning
A	Add the kinship term **after** the relationship phrase

P	Add the kinship term **before** the relationship phrase
blank	Do not add a kinship term

18.5.1 Example phrases

For example:

Adopted > adopting > adopted > A	can produce **adopting father** and **adopted son**

Using P instead:

Adopted > adopting > adopted > P	can produce **father adopting** and **son adopted**. This provides flexibility for languages in which a different word order is appropriate.

Phrase fields can deliberately be left empty:

Birth > > > A	This means that only the kinship term is displayed, such as **father**, **mother**, **son** or **daughter**

.

Anything after the fourth field is ignored by DWE and can therefore be used as a comment:

Birth > > > A > Use the ordinary kinship term for a birth relationship

18.5.2 Default relationship phrases

The following phrases are already built into DWE. These defaults are used even when no Parent-child relationship note has been selected. If you want to change these, you need a note.

Gramps relationship	Parent phrase	Child phrase	Kinship position
Birth	*(blank)*	*(blank)*	A
Adopted	adopting	adopted	A
Stepchild	step	step	A
Sponsored	sponsoring	sponsored	A
Foster	foster	foster	A
Unknown	unknown lineage	unknown lineage	*(blank)*
Custom	custom	custom	A

The optional note does **not** need to replace the entire set of defaults. A line in the note overrides the wording for that relationship. Relationships not mentioned in the note continue to use their built-in defaults.

For example, if the only change required is to put the kinship term before the Adopted phrase, the note only needs:

```
Adopted > adopting > adopted > P
```

There is no need to reproduce all the other defaults.

18.6 Local labels note

The **Local labels note** allows some labels used in the generated website to be replaced with wording chosen by the website creator.

This is useful where different terminology is preferred for a particular family history, or where a more explanatory label is wanted without modifying DWE itself. If you don't need different terminology, you don't need a note.

Select the Gramps note containing the required label definitions. If the option is left empty, the normal DWE labels are used.

18.6.1 Creating a Local Labels note

Create a Gramps note. It can be of any type, although General is probably a useful choice.

Each label replacement is entered on a separate line in the form:

```
existing label > replacement label
```

For example:

```
Why me > Why is this person in the tree?
```

The label on the left must exactly match the existing DWE label. The **>** character separates the existing and replacement labels.

Lines that do not contain the separator are ignored. They can therefore be used for comments or a description of the notes. Putting a description on the first line of the note can also make the note easier to recognise when selecting it from a list of Gramps notes.

In my Local labels note the first line, a comment line, starts with a dash for convenience, so it is sorted towards the top of the note lists when I look for it later.

18.6.2 Selecting the note

When creating the report, select the required note using the **Local labels note** selector on the **DWE Options** page.

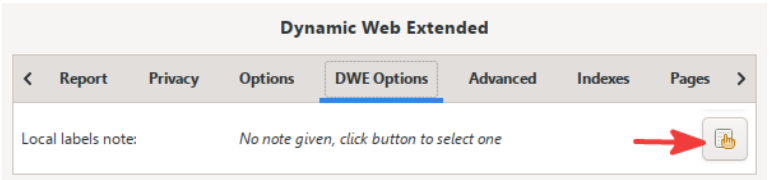


Figure 26- note selector

Generate the report and the specified labels will be replaced.

The changes do not alter the original DWE labels in the DWE code or the information stored in Gramps. To stop using the replacements, clear the **Local labels note** selection and generate the report again.

18.6.3 Scope of a replacement

A Local Labels replacement applies wherever DWE uses the translated label. It is not restricted to a particular page or tab. Care should therefore be taken when replacing a label that may be used in several places.

Local Labels operate on the labels in the language used to generate the report. This means replacements can also be used with translated DWE reports.

18.6.4 Example

For example, DWE uses **Parents & siblings** and **Spouses & children** for the two person-page tabs that the original *Dynamic Web Report* named **Ancestors** and **Descendants**. If you prefer the original terminology, Local Labels can restore it:

```
Parents & siblings > Ancestors
Spouses & children > Descendants
```

The DWE display changes:

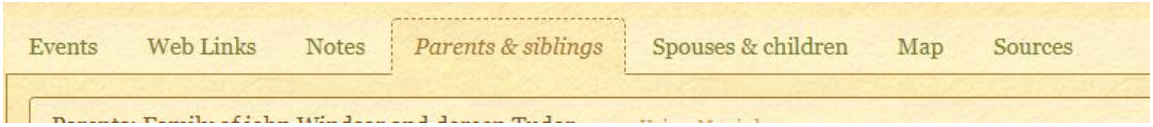


Figure 27- without local labels

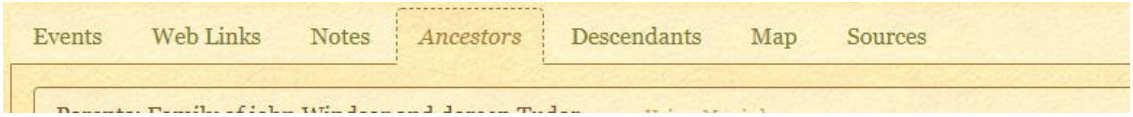


Figure 28- after local labels

18.7 Note types to enhance person display

The next two note types identify a type of note. If a person has a note of that type in their Gramps record, then the content of the note is displayed specially in the top section of their website page. The note types to be identified are the **Why me note type** and **Story note type**.

18.8 Why me note type

The **Why me note type** identifies notes that DWE should put into the **Why Me** information displayed for a person.

18.8.1 Create Why Me note type

You can create a custom note type specifically for this purpose. In my tree I use a custom note type named **Why Me**, but you can use any note type you prefer.

Select that note type in the DWE option for Why me note type. Why me is optional; if no note type is selected, then Why Me is not shown.

18.8.2 Add Why me notes

Prior to generating the DWE report, in your Gramps database you will need to create a Gramps note for a person and attach it to the person. Set the note to the type selected for the Why Me note type.

Attach the note to the person using the Notes tab in their Person Editor. You can share a note between persons if appropriate or create a different note for each person.

18.8.3 Result

DWE finds that note type in a person's data when the report is generated by the creator. The DWE website will display the note in the Why me entry on the person's website page.

See **Why Me** for further explanation.

18.9 Story note type

The **Story note type** identifies the notes used for the short **Story** displayed near the top of a person's page.

The Story note type option is very similar to the Why Me note type option described above. You can create a custom note type specifically for this purpose. In my tree I use a custom note type called **Story**, but you can use any note type you prefer.

A person with a note of that type will then have it displayed as their Story on their website person page.

If no Story note exists for a person, the Story is not displayed. Story is optional. To turn it off, select the blank entry for **Story note type** before generating the report again.

See **Story** for further information.

18.10 Markdown in notes

DWE can render Markdown contained in Gramps notes. There are three ways of identifying notes that should be treated as Markdown. These methods can be used separately or together.

DWE first preserves the normal Gramps styling in a note and then applies Markdown processing. This makes it possible, for example, to add a Markdown table to a note that also contains Gramps formatting such as bold, underline or subscript.

Because this combines two formatting systems, unusual combinations of Gramps styling and Markdown should be checked in the generated report.

18.10.1 Use Markdown note marker

When **Use Markdown note marker** is selected, an individual note can be identified as Markdown by putting:

MD

by itself on the first line.

DWE recognises this as its Markdown marker and while converting the note it removes the marker from the displayed note. The note can be of any type, except Html code and To Do.

The MD marker is a fixed DWE convention. The option determines whether DWE recognises the marker; it does not change the marker itself.

This option is selected initially. It can be turned off if you don't want to use the MD marker method.

18.10.2 Markdown note type

The **Markdown note type** provides another way to identify Markdown notes.

Select a Gramps note type that is to be used for Markdown. Eligible notes of that type are then processed as Markdown without requiring an MD marker on each note.

This can be convenient when Markdown notes are maintained as a distinct type in the Gramps database.

18.10.3 Treat ordinary notes as Markdown

Selecting **Treat ordinary notes as Markdown** applies Markdown processing more generally. Ordinary notes can then contain Markdown without needing either the MD marker or the selected Markdown note type.

The three methods therefore provide different levels of control: the marker can identify individual notes, a note type can identify a group of notes, and this option can apply Markdown processing generally.

Gramps **Html code** and **To Do** notes are excluded from DWE Markdown processing. Their note types take precedence over all three Markdown methods. For example, an Html code note is not processed as Markdown even if its first line contains the MD marker.

18.11 Website themes

The Report options page allows the site creator to affect the appearance of the site using the **Web site template** option. In this guide website templates are called website themes.

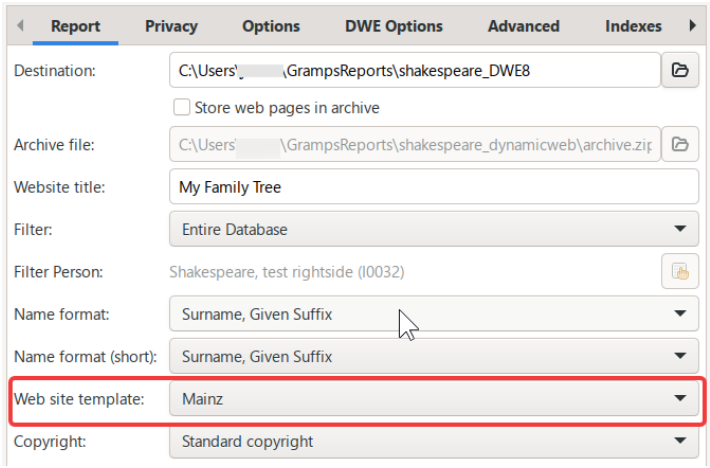


Figure 29- website option

The original DWR provided two options, called Default and Mainz. DWE has increased the number of options. To select a new theme, select its name from the dropdown list.

The theme selected in Gramps determines the initial appearance of the website, but with DWE visitors can subsequently choose a different theme. Their view of the site will change to match their newly selected theme. The visitor option is described in section 16 Website Theme

The new themes are from Bootswatch, a free, open-source collection of themes that change the visual appearance of the Bootstrap front-end framework used inside DWR and DWE. See the Bootswatch site for more information.

18.12 Short name option

DWR provides a **Short Name Format** option in the report options, on the **Report** options page. The short name is already used in the top-level menu Tree when that tree is displayed. DWE also uses the short name in the new one-line heading section of a person page.

19 Program design and development notes

This part of the Guide records some of the design decisions and implementation details behind Dynamic Web Extended. It is intended mainly for anyone maintaining or extending DWE in the future.

DWE began as a modification of the Gramps Dynamic Web Report (DWR). Wherever practical, the existing DWR structure and behaviour have been retained. Changes have generally been made to add a feature, improve the presentation of existing information, or deal with a particular limitation encountered while using DWR.

Some decisions described here are not obvious from the program code itself. They are recorded so that a future change does not inadvertently undo behaviour that was deliberately chosen.

19.1 Relationship data: record rather than infer

DWE tries to report genealogical relationships that are explicitly recorded in Gramps rather than infer additional relationships from the surrounding family structure.

This is particularly important for step relationships. A person should be described as a stepchild when that relationship has been recorded in Gramps. DWE does not assume that a child becomes the stepchild of somebody merely because a birth parent subsequently forms another family.

The same principle is used when identifying siblings. DWE can find possible siblings by following the parent-child relationships recorded in Gramps, including non-birth relationships. It then uses shared birth parents to classify full and half-siblings. If a sibling connection is found but no common birth parent is found, DWE describes the person as a **non-birth brother**, **non-birth sister** or **non-birth sibling**, rather than guessing that the relationship is a step-sibling relationship.

This distinction is deliberate. Family structure alone is not sufficient evidence for DWE to create a genealogical relationship that the researcher has not recorded.

19.2 Parent-child relationship None

An explicit Gramps parent-child relationship of **None** should not be used for relationship traversal. It indicates that the parent-child relationship does not exist or is considered not to exist, and therefore should not be used to discover parents, siblings or other relatives.

Whether an explicitly recorded **None** relationship should nevertheless be displayed to a website visitor is a separate presentation question. This remains an area for possible future work.

19.3 Multiple families and siblings

DWR already supports people belonging to multiple families. DWE makes some changes to the presentation of these families to make less usual family structures easier to follow.

The **Parents & siblings** display associates children with the appropriate parent family and can also show other families of a person's parents. The inherited option controlling this behaviour is labelled **Include children from other families of the parents** in DWE and is selected by default.

This option controls presentation only. It does not alter the genealogy used by other DWE features. For example, clearing the option may hide an additional parent family from **Parents & siblings**, while **Tracking** can still identify a half-sibling and **Events** can still include that sibling's events.

The DWR person-family JavaScript coding was corrected so that parent-family information is matched by the actual family index rather than its position in an intermediate list. This is important when a person has several parent families.

DWE also corrects display problems encountered with families containing children but no recorded parents, and with collapse/expand controls in multiple-family displays. Gramps permits a family containing siblings without requiring an artificial unknown father or mother, and DWE can display such a family correctly.

19.4 Parent-child relationship wording

DWE separates the Gramps relationship type from the wording used to present that relationship. Built-in parent and child phrases are provided for the standard Gramps relationship types, and a **Parent-child relationship note** can override relationship wording.

The optional A and P values allow the kinship word to be placed after or before the relationship phrase. This was designed partly to avoid assuming that English word order will be suitable for every language.

The defaults are always loaded first, and the relationships note then overrides individual entries. A relationship note therefore does not need to reproduce the complete default table.

19.5 Translation and Local Labels

New interface text intended for translation follows the translation mechanism already used by Gramps. DWE-specific strings are added to the translated-label data where they are required by the generated JavaScript.

Local Labels is a separate mechanism. It allows the report creator to replace translated labels without modifying DWE. Replacement occurs after the normal label has been translated, so the label on the left of a Local Labels definition must correspond to the label in the language being used for the report.

A Local Labels replacement is global to that translated label; it is not associated with one particular page or control. For this reason a developer should not reuse the same translated string for unrelated purposes merely because the English wording happens to be identical.

19.6 Topola Tree Viewer

The Tree Viewer is based on Topola Viewer. DWE uses a locally bundled build rather than depending on the online Topola service. The built application is stored under vendor/topola-viewer/dist.

Bundling Topola keeps the generated website self-contained, does not send the family GEDCOM to an external viewer, and avoids cross-site and CORS (Cross-Origin Resource Sharing) issues that can arise when the DWE website and the viewer are hosted at different locations. The Tree Viewer and its supporting files are served from the same website as the rest of the DWE report.

DWE generates the GEDCOM only when **Include Tree Viewer** is selected. The GEDCOM is generated using the Gramps GEDCOM writer and is then sanitised for use by Topola. DWE removes the generated submitter reference, submitter record and generated copyright header so that researcher details are not unnecessarily included in the Tree Viewer data.

DWE's bundled Topola build has also been customised. Among the changes are integration with the DWE person pages, person highlighting, and renaming the export filenames.

Changes to Topola source code are made in the separate Topola source/build directory and rebuilt before the resulting dist files are copied into DWE. Consequently, a future upgrade of Topola must preserve or reapply the DWE modifications rather than simply replacing the bundled files with a new upstream build.

19.7 Tree Viewer limitation

Tree Viewer may fail to render large trees. It appears that Topola is in the process of developing expandable tree controls. This is a path for further investigation. If the Viewer could initially display a

limited number of generations and expand the tree on demand, it might avoid the need to render a very large tree at once.

19.8 Lifespans and Vega

The **Lifespans** display uses Vega rather than the normal DWR charting code.

DWR generally exports small individual categories of information into separate JavaScript data structures and combines them in the browser. The Lifespans implementation takes a somewhat different approach: much of the information needed to construct a person's timeline row is assembled together for the chart. This was a deliberate implementation choice because the timeline needs several related values at the same time.

Lifespans is person-oriented rather than family-oriented. The chart gathers the subject's parents, siblings, spouses and children and arranges them into those groups. People are de-duplicated where necessary.

Vega export is configured to use lifespans as the download filename and a white chart background, so exported SVG and PNG files have predictable filenames and appearance.

19.9 Markdown processing

DWE supports three ways of identifying Markdown: the fixed first-line MD marker, a selected Markdown note type, and the **Treat ordinary notes as Markdown** option.

MD is deliberately a fixed marker rather than a configurable string, making marked notes portable between DWE installations and family trees.

Html code and **To Do** notes always take precedence over the Markdown mechanisms and are excluded from Markdown processing.

DWE preserves Gramps styled-note formatting and then allows Markdown processing of the resulting content. This permits useful combinations such as a Markdown table within an otherwise normally styled Gramps note. Because Gramps styling and Markdown are two independent formatting systems, unusual combinations may still need to be checked in the generated website.

19.10 Gramps styled-note compatibility

Gramps 5.2.3's HTML backend does not provide mappings for the Strikethrough and Subscript StyledText types. As a result, the original DWR can fail when it encounters notes containing those styles.

DWE uses its own small extension of the Gramps HTML backend to provide HTML mappings for these styles (<s> and <sub>). This retains compatibility with Gramps 5.2 while allowing the normal Gramps implementation to handle the other StyledText types.

19.11 Participation events labels

Participation event labels are two-line labels displayed like this: Marriage followed by (:Best man) for a person with a ‘Best man’ role. The colon identifies participation roles, in case parentheses may be part of an event type name, such as Marriage (Christian).

19.12 Diagnostic logging

DWE uses the Gramps/Python logging system for development diagnostics.

Normal diagnostic messages should use DEBUG level so that they do not interfere with ordinary Gramps operation. WARNING and particularly ERROR should be reserved for conditions that the user needs to know about; Gramps may present ERROR messages through its error-reporting interface.

Logging output appears in the console when Gramps is started from the console.

19.13 Development and source control

DWE is maintained as a Git repository. Development changes should normally be made in a separate branch, tested by generating and viewing a report, and committed in small, logical checkpoints.

Generated websites should be tested through a local HTTP server rather than solely with file:// URLs. Some browser features, including the Tree Viewer, require normal web-server behaviour. DWE provides **View report locally.bat** for convenient local viewing on Windows.

Changes involving bundled third-party components such as Topola should distinguish between changes to the upstream source and the built files stored in the DWE repository.

19.14 Browser storage

Some DWE display preferences are saved in the browser so that they can be remembered as the visitor moves between pages or returns to the website later.

Web browsers provide two similar forms of storage that can be used for this purpose:

Session storage retains information for the current page session. It can carry a setting between pages while that session continues, but is not intended to persist as a long-term preference.

Local storage persists beyond the current browser session. It allows a website to remember a preference when the visitor closes the website and returns to it later.

DWE uses browser storage for some visitor display preferences rather than storing them in the generated family-history data. These settings therefore belong to the browser being used, not to the Gramps family tree or the DWE report options.

Browser storage must be permitted for these preferences to be remembered. Most browsers allow it by default. Private or incognito browsing, clearing site data, or privacy settings that remove or prevent site storage may cause preferences to be lost.

These browser-stored preferences should not be confused with the **report options** used when generating DWE. Report options are saved in a file by Gramps for the family tree; browser preferences are choices made while viewing the generated website.

19.15 Additional CSS and JavaScript

DWE retains the CSS and JavaScript structure of the original *Dynamic Web Report* and adds or modifies files to support its additional features.

The `dwr_start.js` file loads the JavaScript and style sheets used by the generated website. In addition to the standard DWR files, DWE loads `dwr_styles_B.css`, which contains additional DWE styling, and `timeline-vega.js`, which contains JavaScript used for the Lifespans display.

Much of the DWE website behaviour is implemented in the existing DWR JavaScript structure rather than in separate files for every DWE feature. For example, DWE additions to person pages and other displays are integrated with the existing `dwr.js` processing.

Keeping the additional Lifespans code in `timeline-vega.js` helps separate the Vega-based graphical display from the larger `dwr.js` file. Additional DWE CSS in `dwr_styles_B.css` similarly allows DWE-specific presentation changes to be kept separate from the main DWR style sheet.

19.16 Generated website data files

DWE reuses the architecture of the original *Dynamic Web Report* in which much of the genealogical information used by the website is exported into JavaScript data files. The browser uses this data to construct person, family and other displays dynamically rather than requiring a separately generated HTML page containing all the information for each person.

DynamicWebExtended.py exports the principal Gramps objects into JavaScript data structures. These include individuals, families, places, sources, citations, repositories and media. References between objects are generally stored as indexes into the corresponding data structures; for example, an event can contain a place index and citation indexes rather than repeating the complete place and citation records.

DWE extends some of the data inherited from DWR to support its additional features. Person data, for example, includes information used for Lifespan, Why Me, Tracking and Story, as well as the existing person and event information.

Tree Viewer is somewhat different. When **Include Tree Viewer** is selected, DWE also generates a GEDCOM file for the bundled Topola viewer. This is separate from the JavaScript data used by the normal DWE pages. Before the GEDCOM is included in the generated website, DWE removes the Gramps researcher/submitter information and the automatically generated copyright statement.

The generated website therefore contains both the program files that provide the browser interface and generated data representing the selected family tree. These files belong together: when a generated DWE website is copied to another location or uploaded to a web server, the complete generated directory should be copied.

The generated data filenames reflect these object types. I represents individuals (persons), F families, P places, S sources, C citations, M media and R repositories. The data is divided further into files containing particular fields. For example, `dwr_db_I_birth_date_0.js` contains extracted birth-date data for individuals. If the data for a field is sufficiently large, it is split across a series of numbered files, such as `..._0.js`, `..._1.js`, `..._2.js` and so on.

19.17 Known issues and future work

Areas identified for possible future work include:

- **None relationships** - deciding whether explicitly recorded **None** parent-child relationships should optionally be displayed, while continuing to exclude them from relationship traversal;
- **Help page in DWE report options dialog** - reviewing the DWE help-page destination rather than retaining the original DWR help link. The **help page** link is hard coded inside Python – the existing report points it to the original Dynamic Web report help page. I would need to adapt that at some time and point this to its own page;
- **Hyperlink displa**y- reviewing the visual treatment of hyperlinks where themes do not make clickable text sufficiently obvious;
- **Version testing** - testing DWE against future Gramps versions in an isolated environment before declaring them supported;
- **Local server window** - automatically closing of the local server command window after a user closes the DWE browser page;
- **Optimise media file copying** - improving the `copy_file()` function to avoid unnecessary copying of unchanged files. Currently, DWE (and DWR) copies each source file to a temporary destination

and then compares its contents with the existing file, even when the files are identical. Consider using file size, modification timestamps or other methods to identify unchanged files before copying, while ensuring that modified files are reliably detected. This could significantly reduce report generation time for websites containing large numbers of media files, particularly when regenerating an existing website.

20 Report generation progress and timing

When you click **OK** to generate a report, *Dynamic Web Extended* (DWE) displays a progress dialog showing the stages of report generation and the overall percentage completed.

DWE also measures the elapsed time for each stage. These timings are written to the console and can be viewed if Gramps was started from a command-line console. The timing information is primarily intended for technical diagnosis and performance analysis.

20.1 Understanding the progress indicator

The amount of processing required varies considerably between family trees. For example, a tree containing many media files may spend substantial time copying images and generating thumbnails, while another tree may contain thousands of sources or citations.

The progress indicator does not analyse those differences; it just uses fixed estimates for the different stages. Consequently, the progress bar may advance quickly during some stages and more slowly during others. The percentage displayed is only an estimate of the work completed.

During particularly long operations, Windows may display a message indicating that **"Gramps is not responding"**. This can occur when Gramps is busy processing data and is temporarily unable to respond to Windows. The message will often disappear when processing continues. It does not necessarily indicate that Gramps has stopped working.

20.2 Example timing output

The following example shows the timing information produced during the generation of a website:

- DWE TIMING: Building object dictionaries...: 4.583 seconds
- DWE TIMING: Sorting object dictionaries...: 0.052 seconds
- DWE TIMING: Preparing website files...: 0.141 seconds
- DWE TIMING: Exporting individuals...: 15.080 seconds
- DWE TIMING: Exporting families...: 0.373 seconds
- DWE TIMING: Exporting sources...: 8.333 seconds
- DWE TIMING: Exporting citations...: 0.365 seconds
- DWE TIMING: Exporting repositories...: 0.006 seconds
- DWE TIMING: Exporting places...: 0.330 seconds
- DWE TIMING: Exporting media...: 16.462 seconds
- DWE TIMING: Building surname index...: 0.157 seconds
- DWE TIMING: Generating HTML pages...: 1.402 seconds
- DWE TIMING: Generating GENDEX index...: 0.035 seconds
- DWE TIMING: Generating GEDCOM for Tree Viewer...: 3.725 seconds
- DWE TIMING: Preparing local viewing files...: 0.003 seconds
- DWE TIMING: Creating website archive...: 37.773 seconds
- DWE TIMING: Total report generation: 89.289 seconds

In this example, exporting individuals, media and sources accounted for much of the processing time, though building an archive at the end took the most time. Some stages, such as website archiving, are optional. If not enabled, the timing is still shown.

The actual timings depend on the size and contents of the family tree, the report options selected, and the performance of the computer and its storage devices.

21 Appendix

21.1 Example Gramps notes

The following examples provide starting points for some of the special notes that can be used with DWE. They can be copied into Gramps notes and adapted for a particular website.

All these notes are optional. The relevant sections of this guide describe how each note is selected and used.

The Events, Lifespans and Tree Viewer note examples below are display notes and use Markdown for convenient formatting. The Parent-child relationship and Local labels notes are text notes used inside DWE: DWE reads their contents directly, so they should follow the syntax shown in their examples and should not be formatted as Markdown.

21.2 Example Events About note

This example note uses DWE's **MD marker** to render the note contents as Markdown. The **MD** line is not displayed on the website.

MD

Events

The events list has been extended, and brings together events for the person and their family.

You can choose to display:

- the person's own events
- events for their parents
- events for their siblings
- events for their spouses
- events for their children
- events in which the person has a role (called Participation events)

Events are normally shown in date order. Events without dates cannot be positioned chronologically and are not included when the display is limited to the person's lifespan.

Use the checkboxes above the table to choose the information you want to see. Your display choices are remembered as you move between people in the website.

Event notes can also be shown or hidden. Hiding notes can provide a more compact view when many events are displayed.

21.3 Example Lifespans About note

This example note uses DWE's **MD marker** to render the note contents as Markdown. The **MD** line is not displayed on the website.

```
MD

## Lifespans


This chart shows the lifespans of the person and their immediate family, grouped as parents, siblings, spouses and children. Marriages are shown as vertical markers.


Select a person's name to open their page in this website.


Use **Date hints** to show or hide date markers, and **Marriages** to show or hide marriage events. The **Label height** control changes the size of the labels.


The three-dot menu at the top right of the chart can be used to save the chart as an SVG or PNG image.
```

21.4 Example Tree Viewer About note

This example note uses DWE's **MD marker** to render the note contents as Markdown. The **MD** line is not displayed on the website.

```
MD

## Tree Viewer


The Tree Viewer provides an interactive view of the family tree centred on the current person.


You can pan and zoom the tree, search for another person, and select people to see more information about them.


You can maximise the tree to use the whole browser space.


Select **Open page for this person** to open the selected person's page in this website.


The View options allow different tree layouts to be selected. The Download options can save the displayed tree as PDF, PNG or SVG.
```

21.5 Example Parent-child relationship note

DWE already contains definitions for the standard Gramps parent-child relationships, so there is no need to copy all the defaults into this note. Include only the relationships whose wording you want to change.

This is not a Markdown note. DWE reads new relationship definitions directly from the note.

See **Parent-child relationship note** in **section 18.5** for the complete syntax and the built-in defaults.

```
Alternative parent-child relationship phrases

Lines without a > separator can be used as comments.


Adopted > adopting > adopted > A > can produce ‘adopting father’ and ‘adopted son’

Foster > foster > foster > A
```

21.6 Example Local labels note

This example shows how to restore the original DWR wording for Ancestors and Descendants, plus a new label for **Why Me**.

The Local Labels note **is not a Markdown note**. DWE reads each label replacement directly from the note. See the Local Labels section 18.6 for the complete syntax.

```
Local label changes


Parents & siblings > Ancestors

Spouses & children > Descendants

Why me > Why is this person in the tree?
```

(end)